

JKBOSE 2026

CLASS 11TH

Bhaiichara
ON
TOP

#XUVV♥ ORGANIC CHEMISTRY: SOME BASIC PRINCIPLES AND TECHNIQUES

LECTURE-07 CHEMISTRY

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Topics To Covered

1

Nomenclature of Functional Groups Part 02

REVISION



1) ALKANES, ALKENES, ALKYNES (CYCLO).

2) Functional groups.

— COOH — oic acid

— COOR — oate

— COX — oylhalid

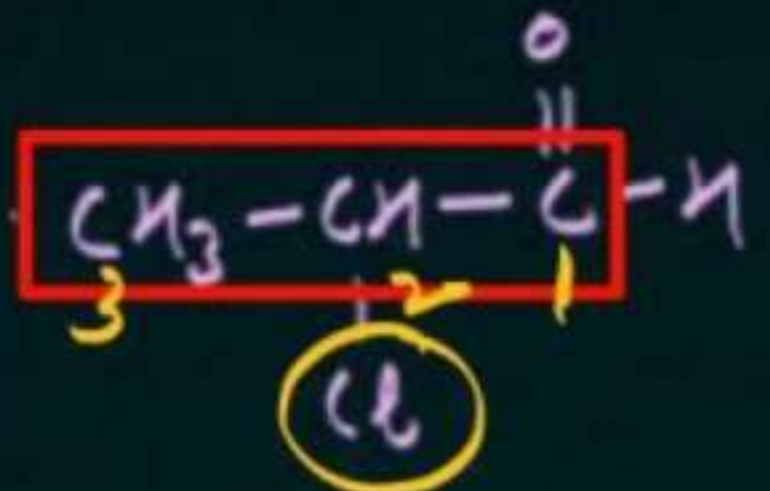
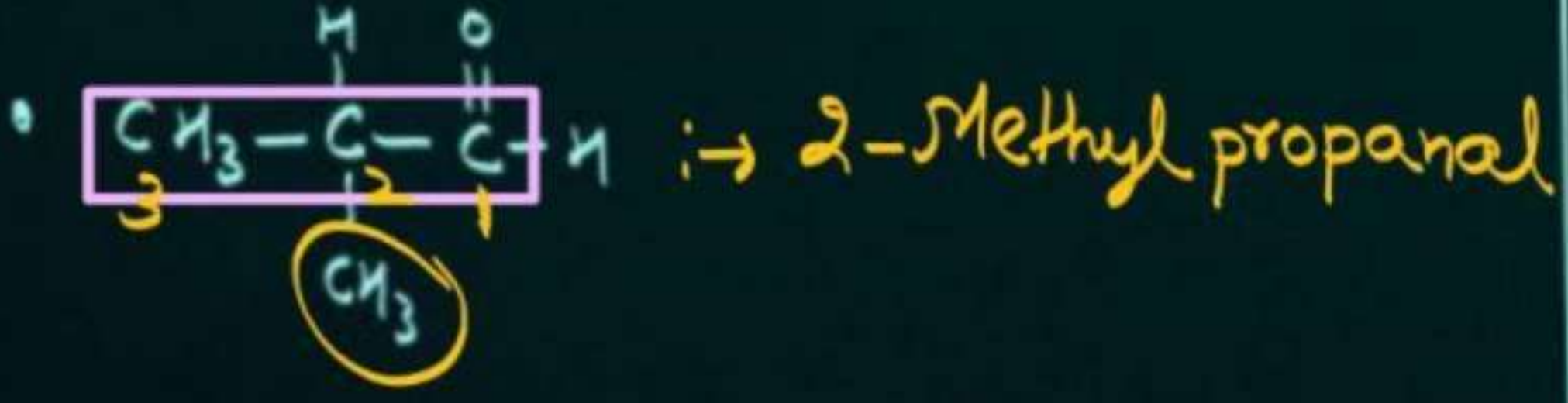
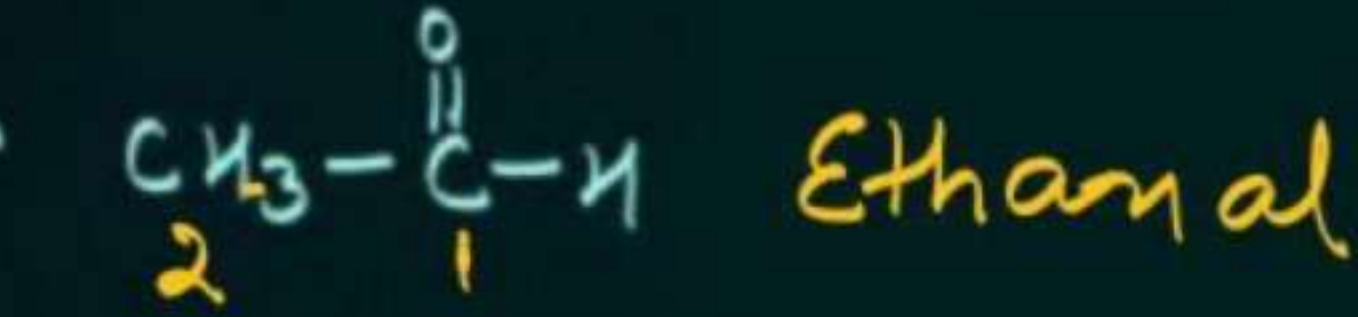
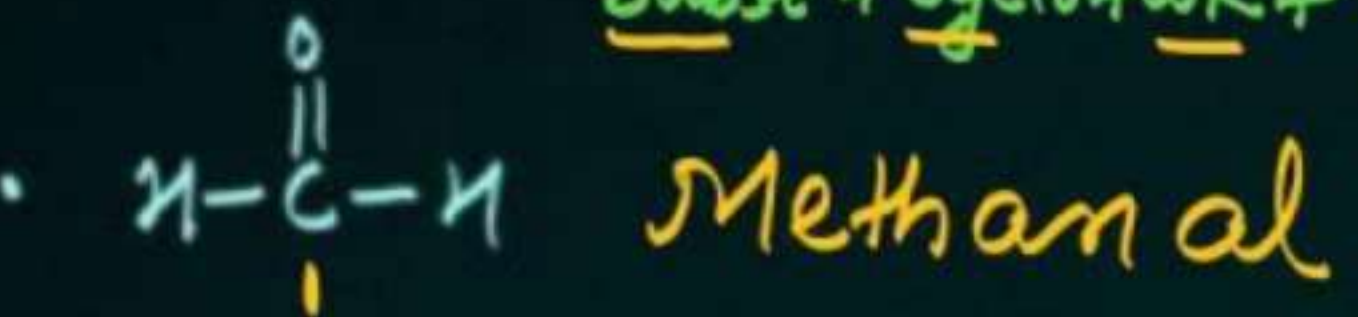
— CONH_2 — Amide

— CN — nitrile

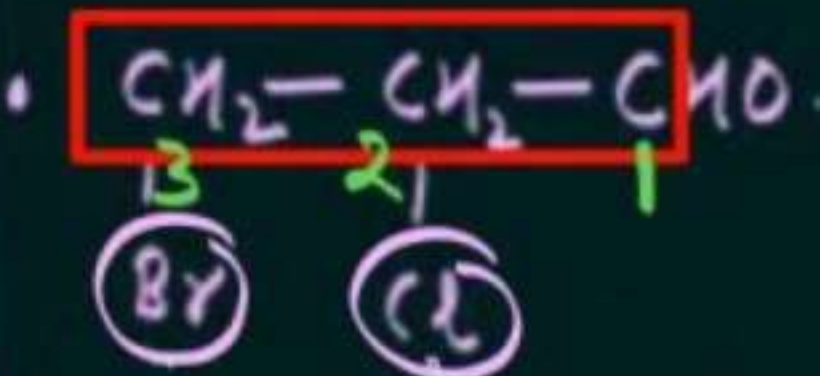
— $\text{—}\overset{\text{O}}{\parallel}\text{C—H}$ — al.

6) Aldehydes ($-\overset{\overset{\text{O}}{\parallel}}{\text{C}}-\text{H}$). or $-\text{CHO}$

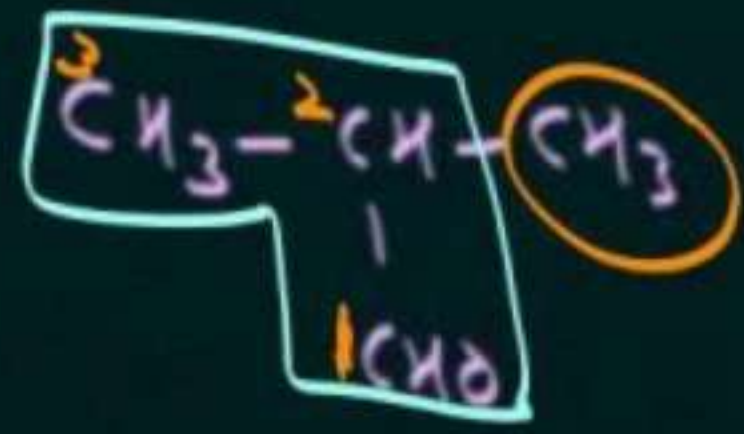
2^o- Suffix Used = al.
 Subst + cyclo + WR + ene or ane + al



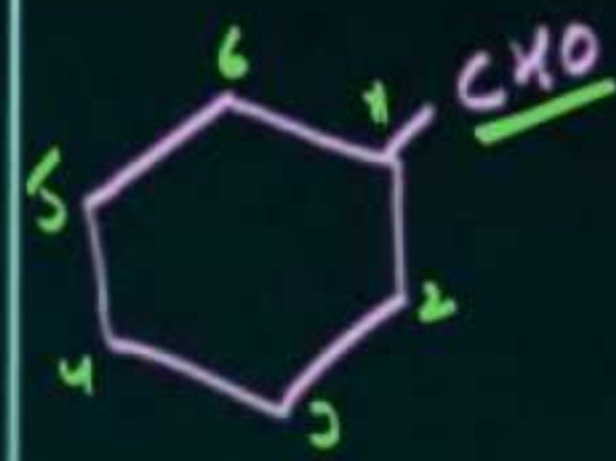
2-Chloro propanal



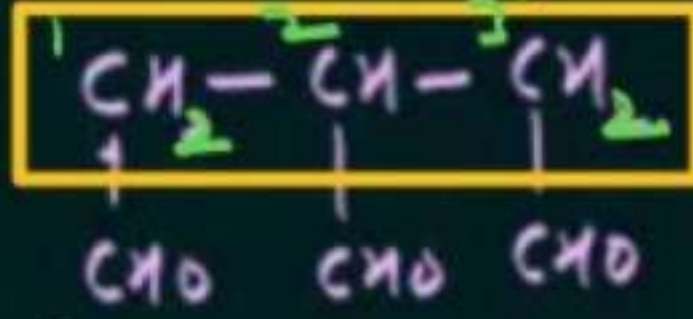
3-Bromo-2-chloro propanal



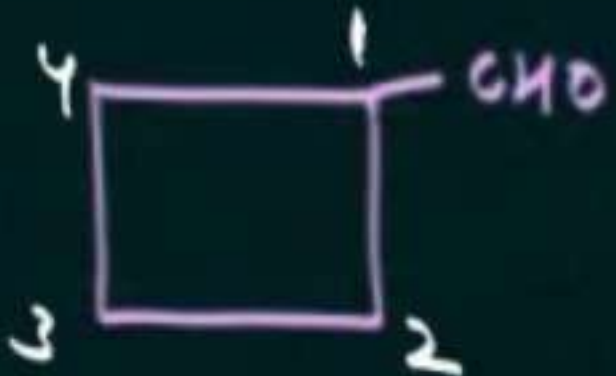
2-Methyl propanal



Cyclohexane carbaldehyde



Propan-1,2,3-tricarbaldehyde



Cyclobutane carbaldehyde



Benzene carbaldehyde or Benzaldehyde.

If -CHO directly on ring or in case of symm. molecule if we cant. include any one of -CHO in pic then we exclude each -CHO & use 2^o-suffix 'carbaldehyde'.

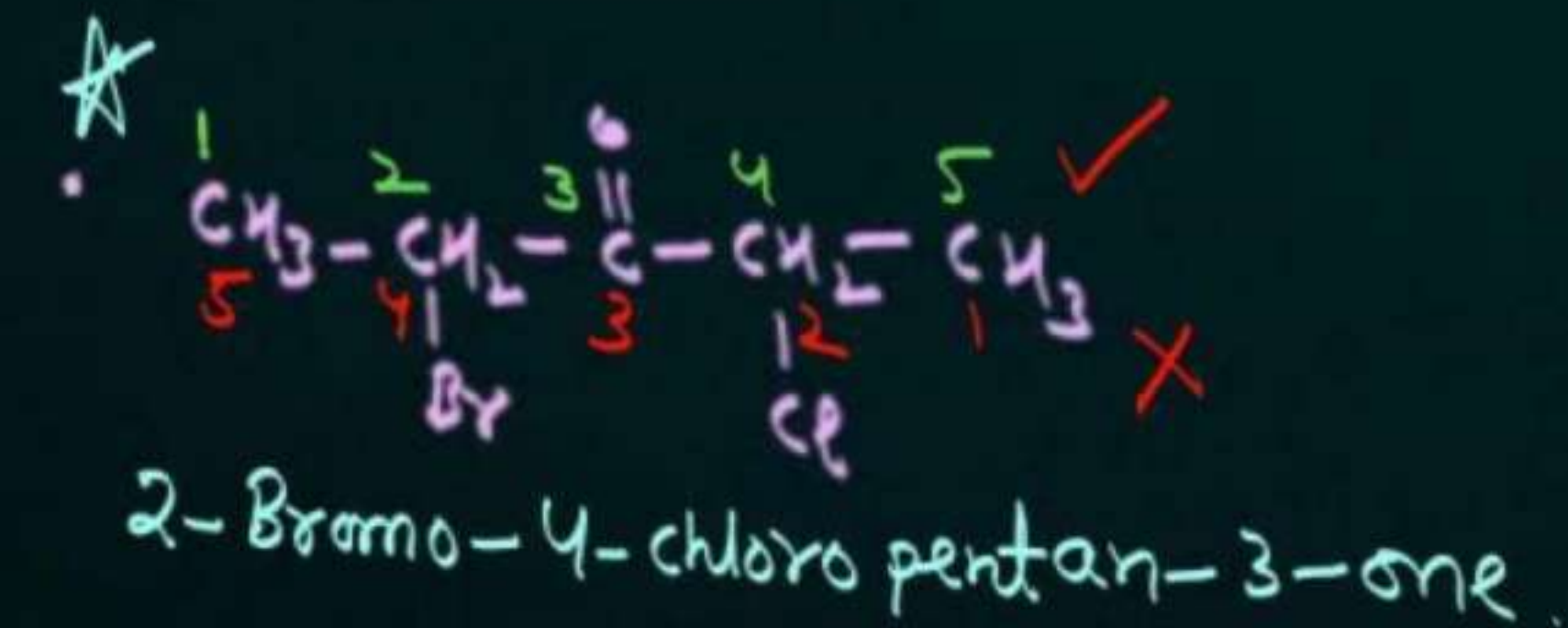
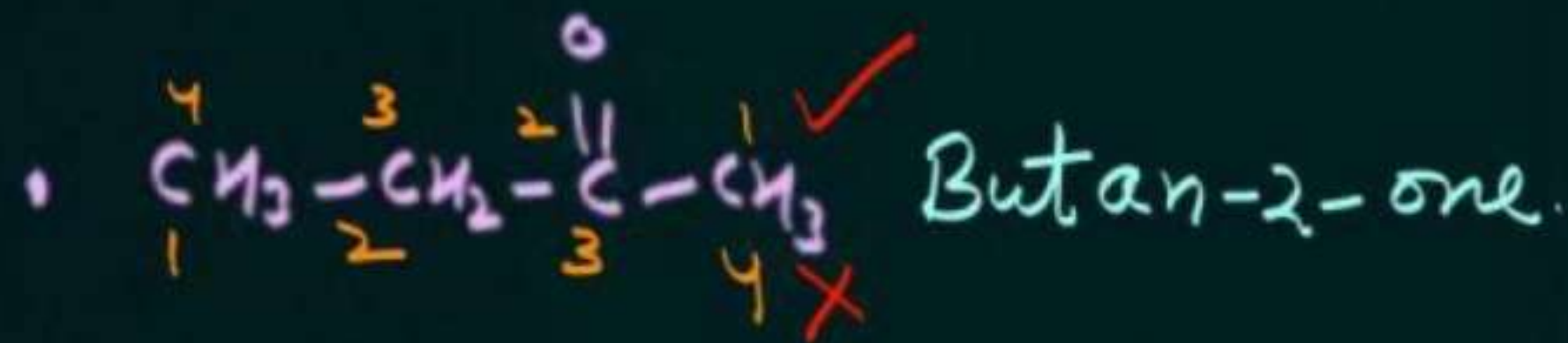
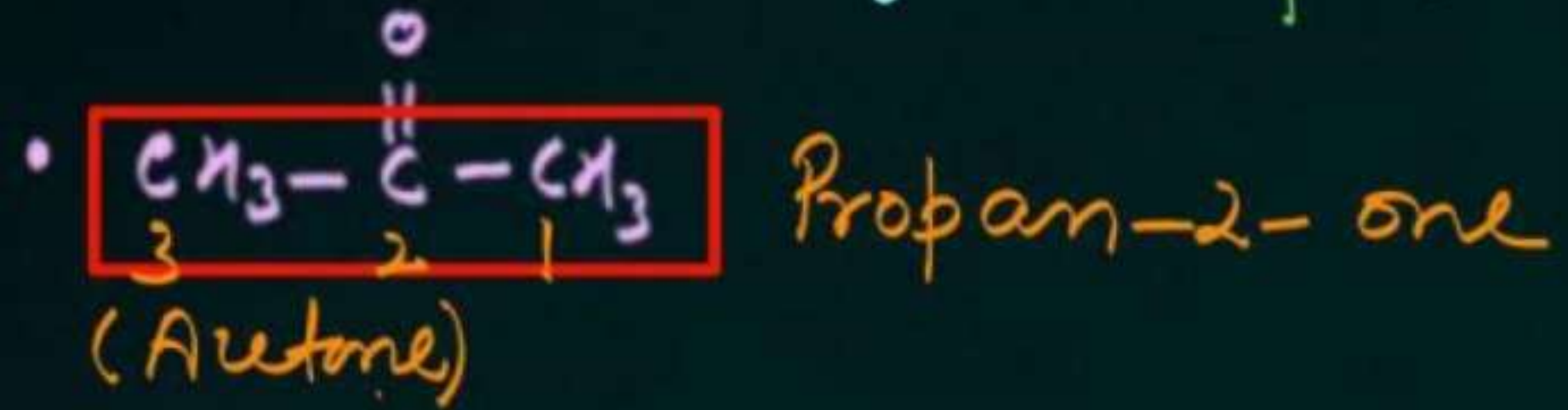
1) Decoding the PCC $\Rightarrow$ $\begin{matrix} \text{Max. No.} \\ \text{of FG} \end{matrix} \begin{matrix} \text{Max. no. of} \\ \text{MB} \\ (=1\equiv) \end{matrix} \begin{matrix} \text{Max. no.} \\ \text{of} \\ \text{C-atoms} \end{matrix} \begin{matrix} \text{Max. no.} \\ \text{of} \\ \text{Substituents} \end{matrix} \text{Alphabetical order}$

2) Numbering the PCC $\Rightarrow$ $\begin{matrix} \text{Lead no.} \\ \text{FG} \end{matrix} \begin{matrix} \text{MB} \\ (=1\equiv) \end{matrix} \text{Substituent} \text{Alphabetical order.}$

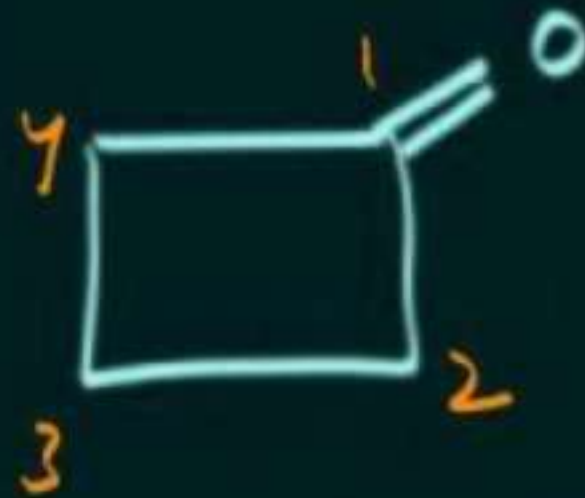
3) Naming $\Rightarrow$ SUBSTITUENT + CYCLO + WR + $\begin{matrix} \text{ane} \\ \text{or} \\ \text{ene} \\ \text{or} \\ \text{yne} \end{matrix}$ + FG.

7) KETONE ($R-\overset{\overset{O}{\parallel}}{C}-R$).

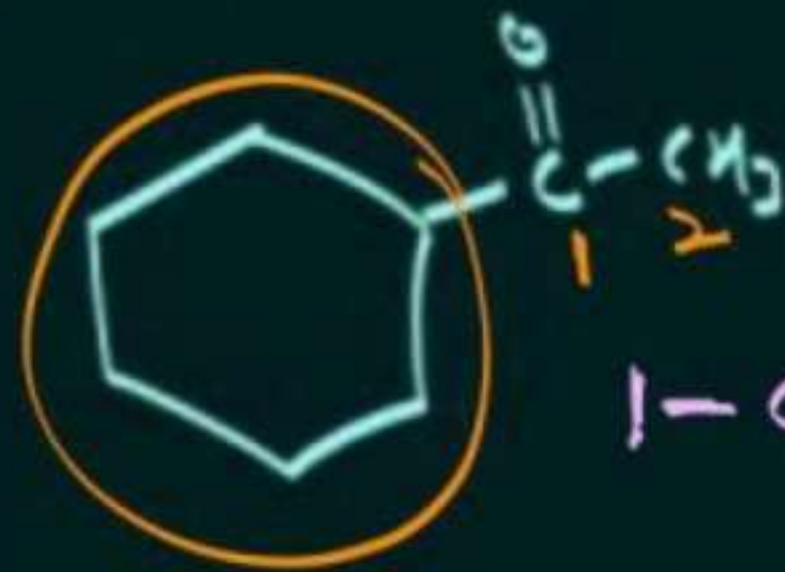
2°- Suffix used = one.
 Subst + cyclo + WR + ^{yl} or ^{ane} + one



Cyclohexan-1-one



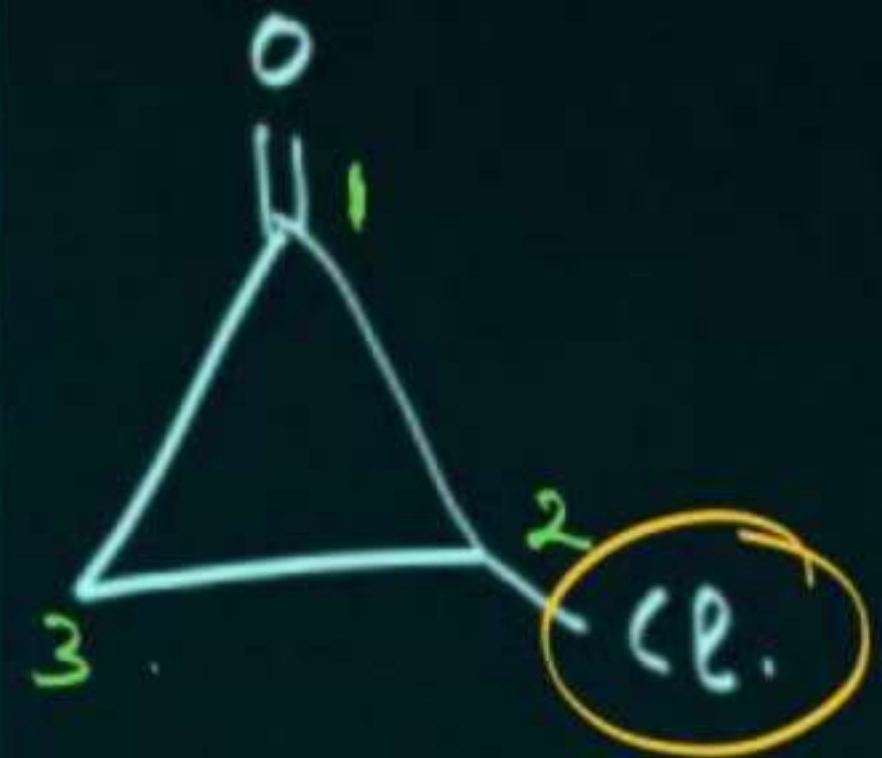
Cyclobutan-1-one



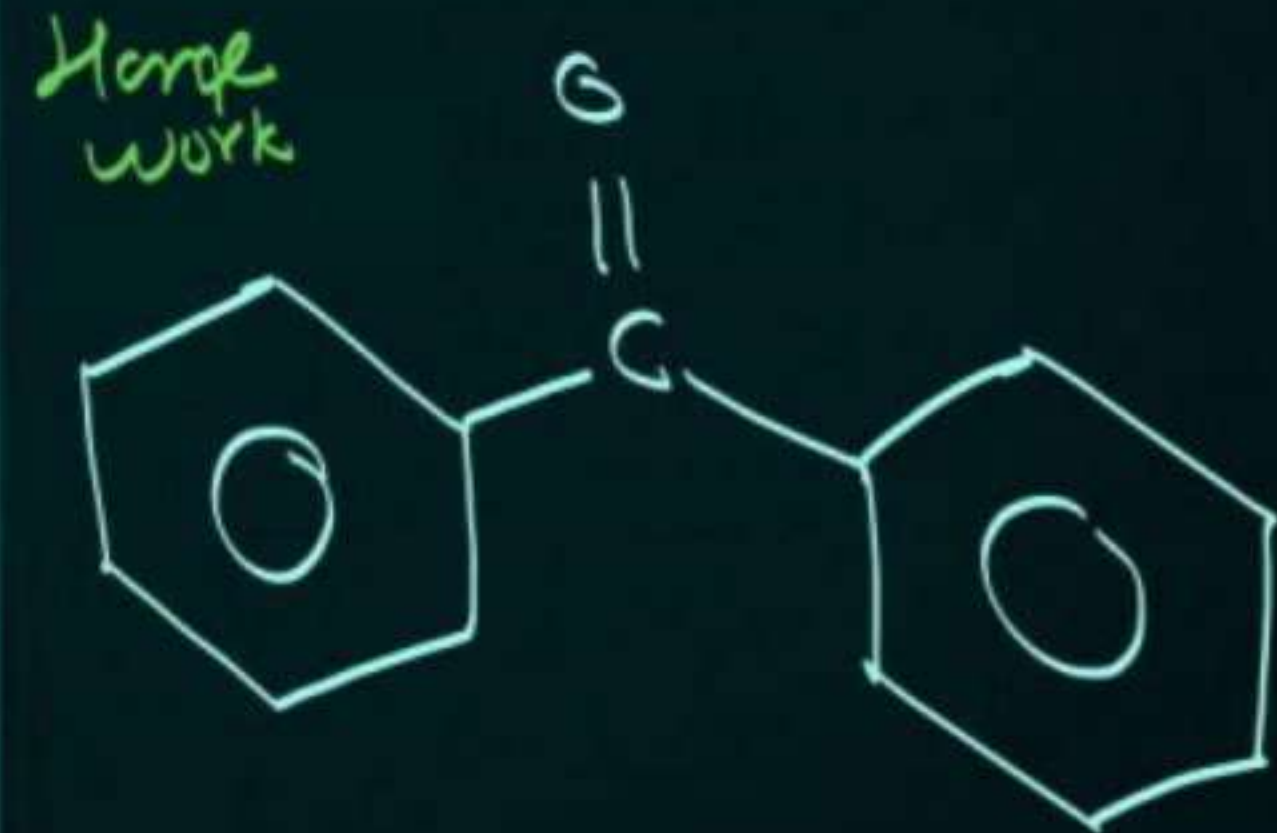
1-cyclohexylethanone



1-Phenylethanone



2-Chlorocyclopropan-1-one



8) Alcohols $\rightarrow (-OH)$

-SH (Thiol)
2°-suffix = Thiol

2°-suffix = ol

Subst. + Cyclo + WR + ene or yne + ol

• $CH_3 - \underline{OH}$ Methanol

• $CH_3 - \underline{CH_2} - OH$ Ethanol

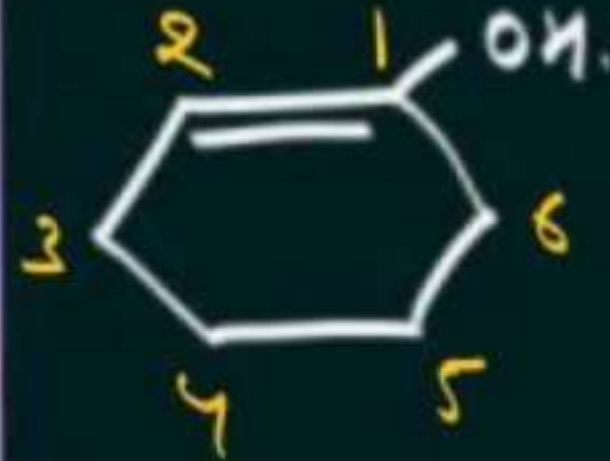
• $CH_3 - \underline{CH_2} - OH$ 1-Bromoethan-1-ol
Br



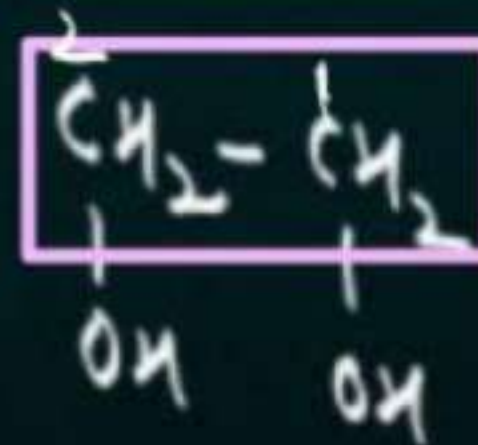
But-3-en-1-ol



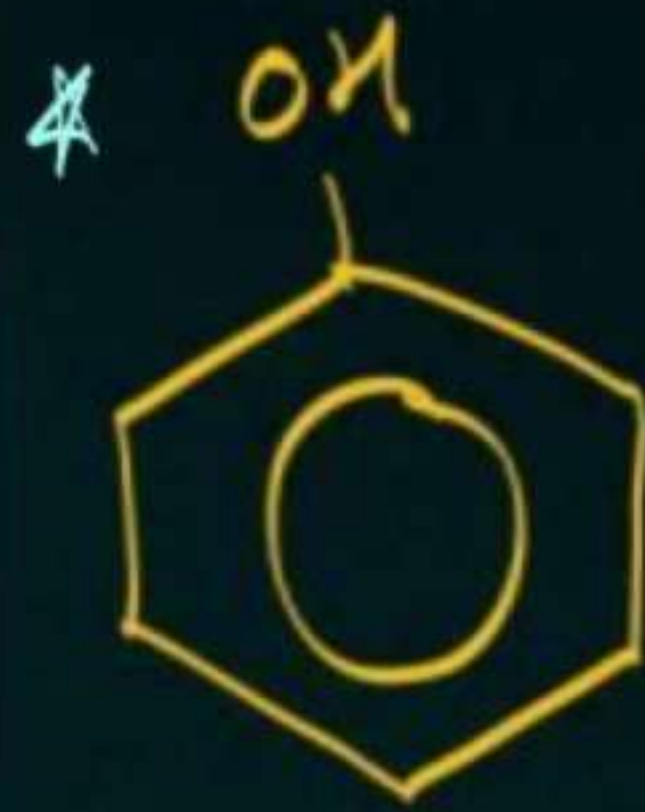
Cyclobutan-1-ol



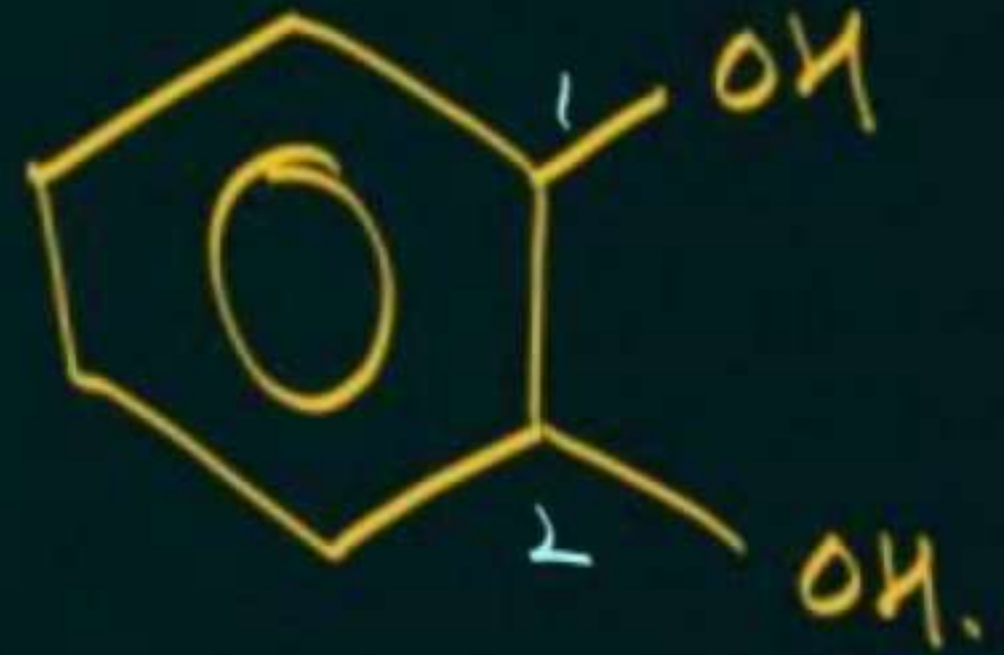
Cyclohex-1-en-1-ol



Ethane-1,2-diol



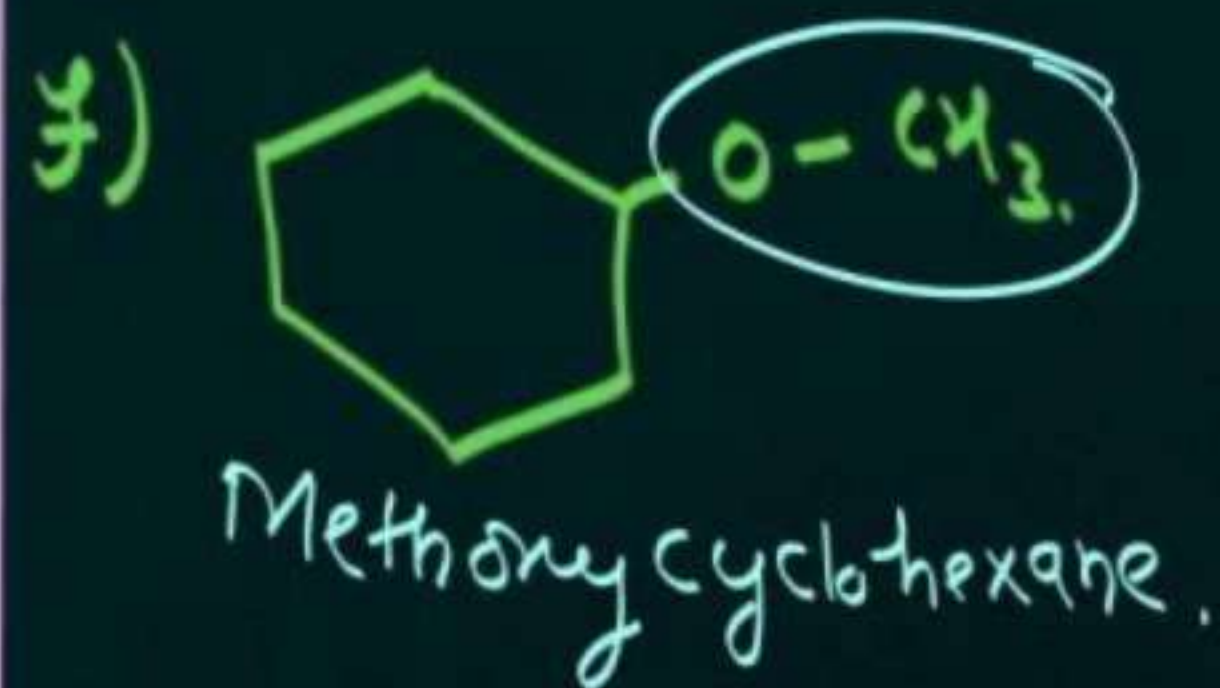
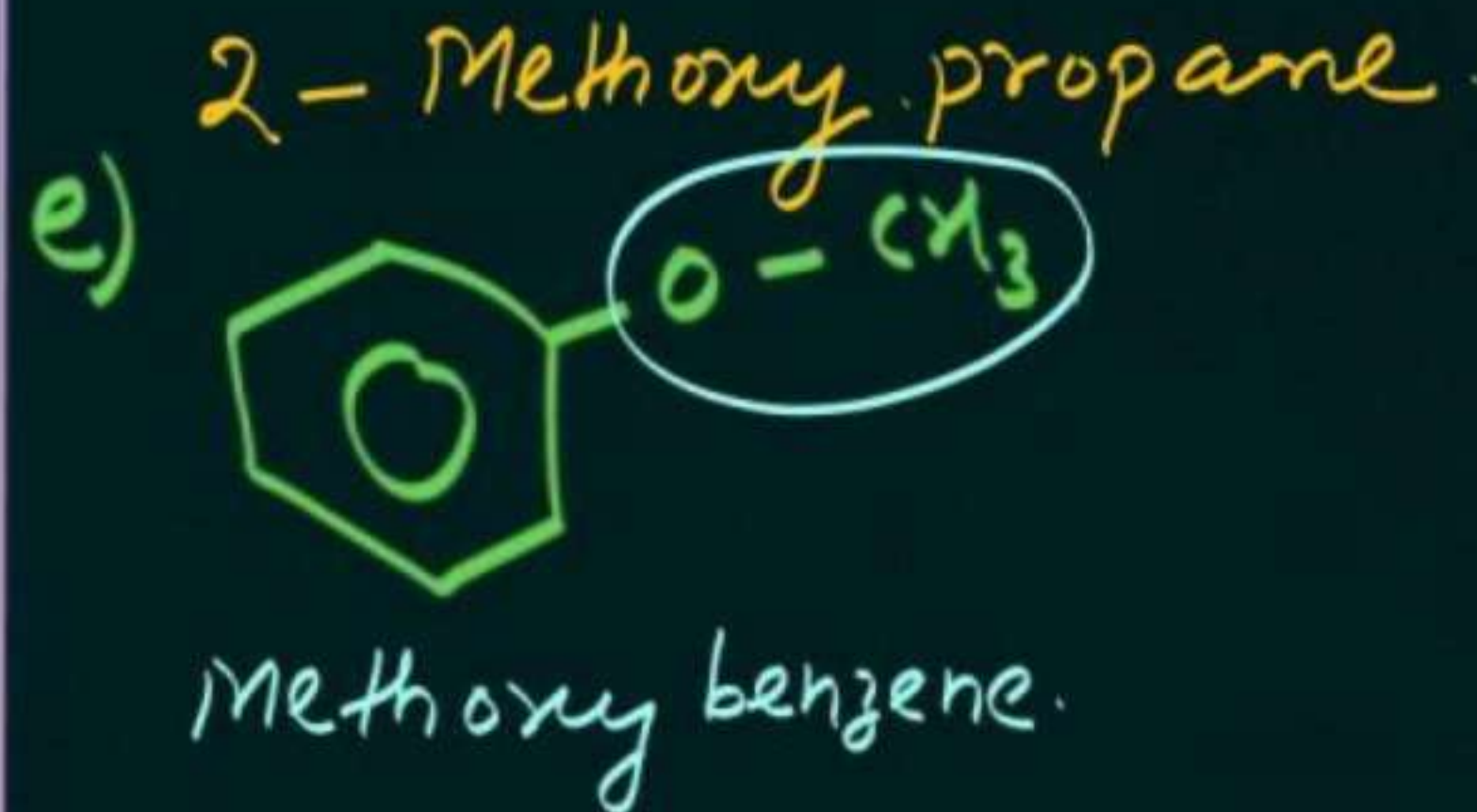
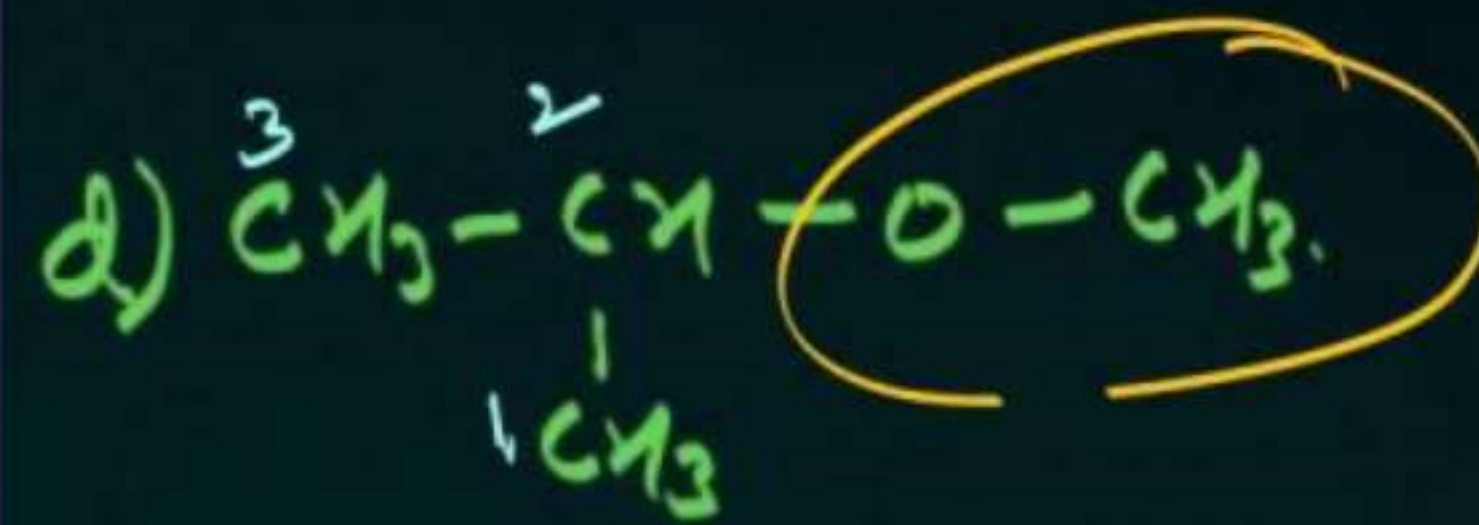
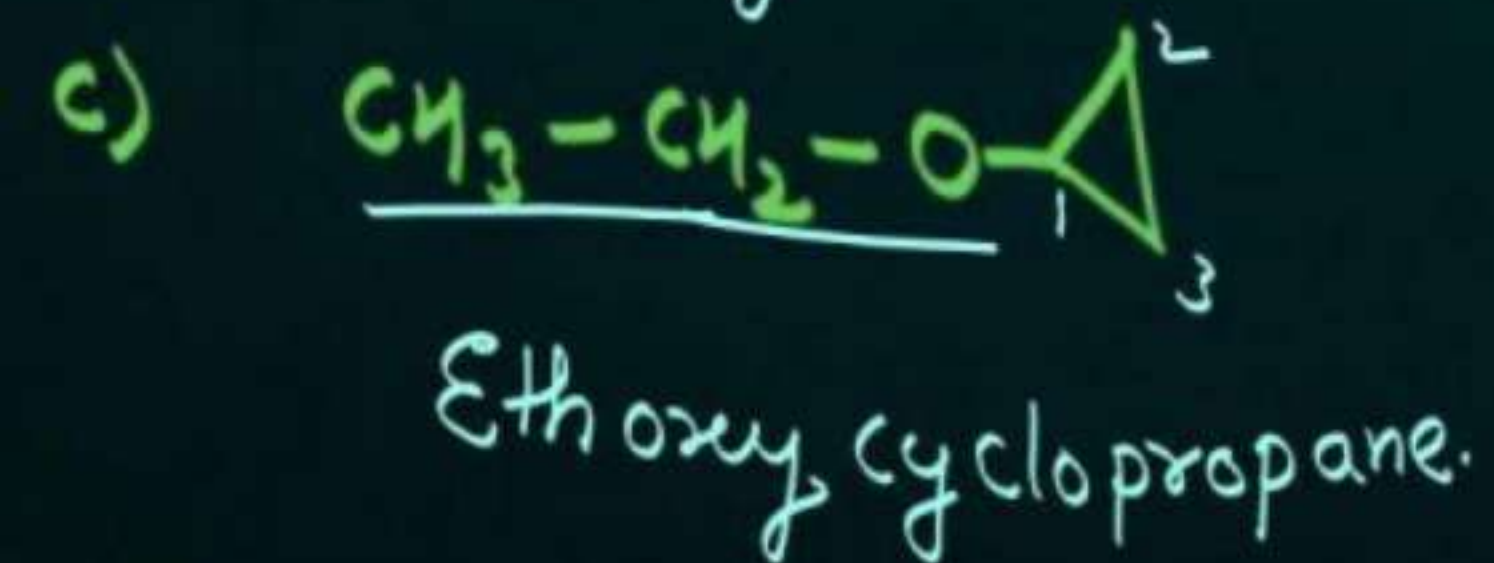
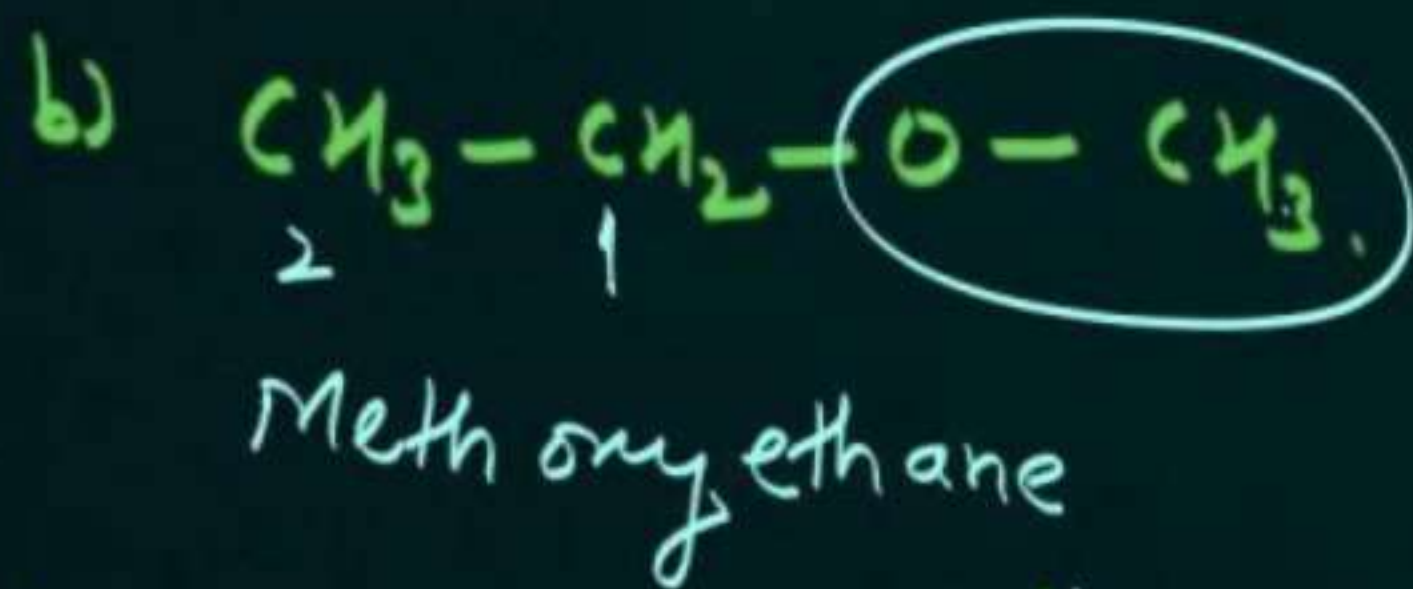
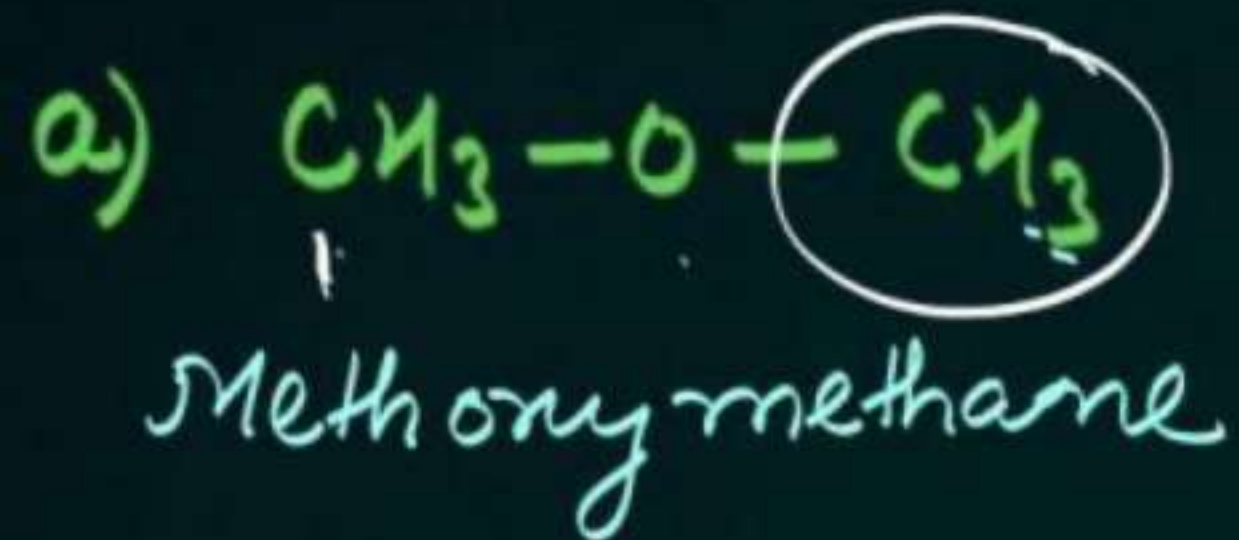
PHENOL



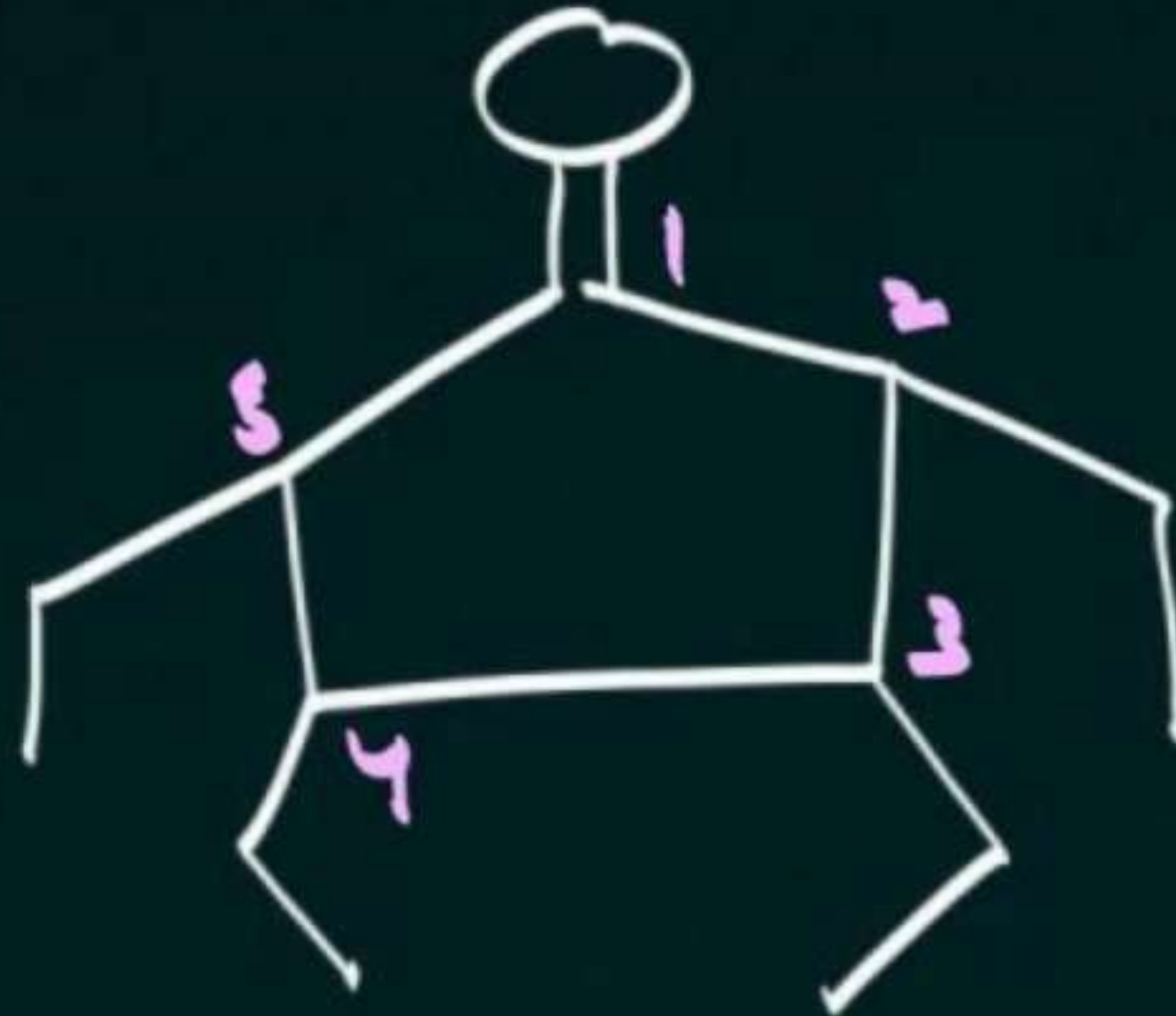
Benzene-1,2-diol

10) ETHERS (R-O-R)

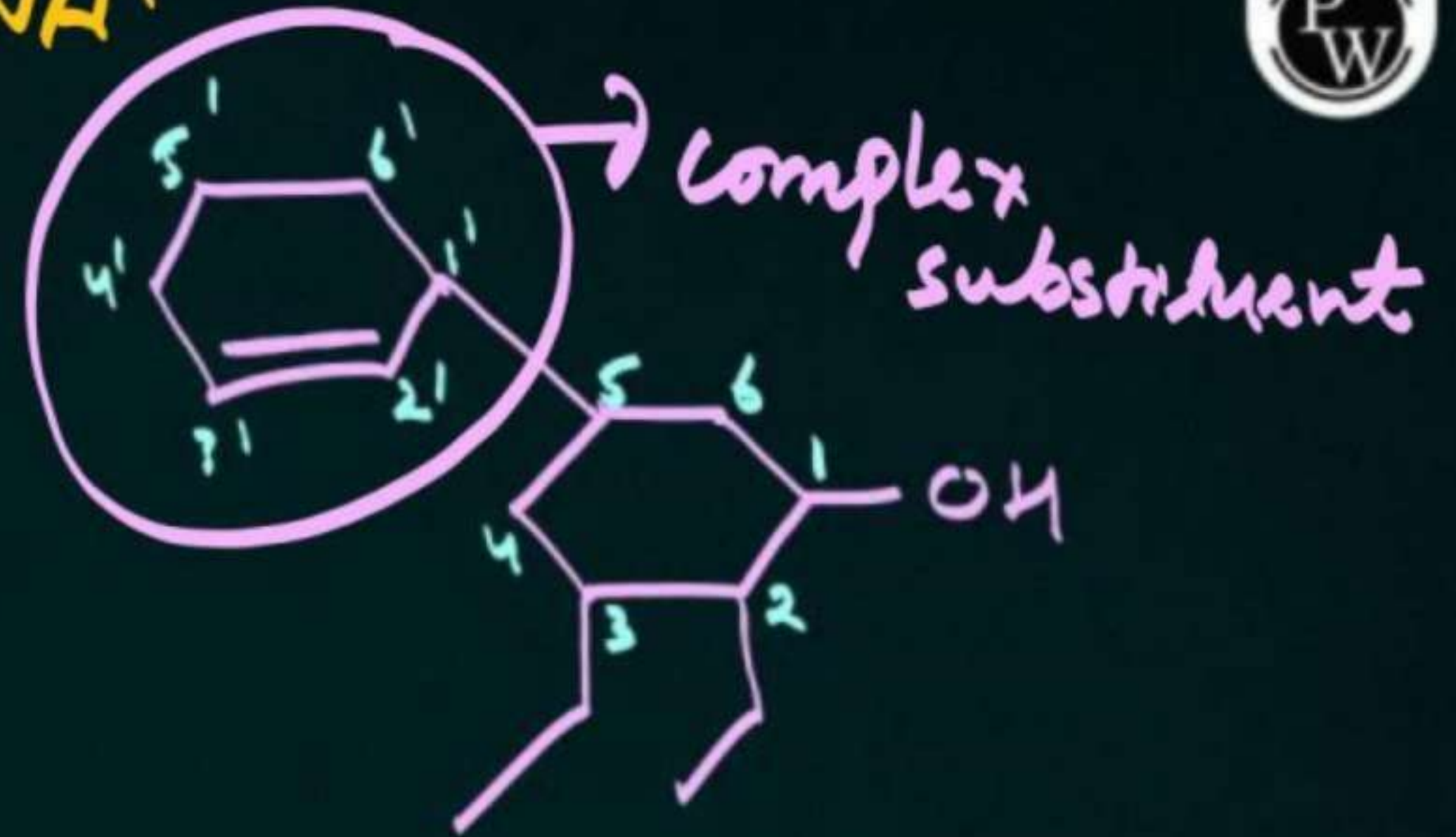
Acts as substituent.



PRACTICE → DUM NA TO KAR K LANA.



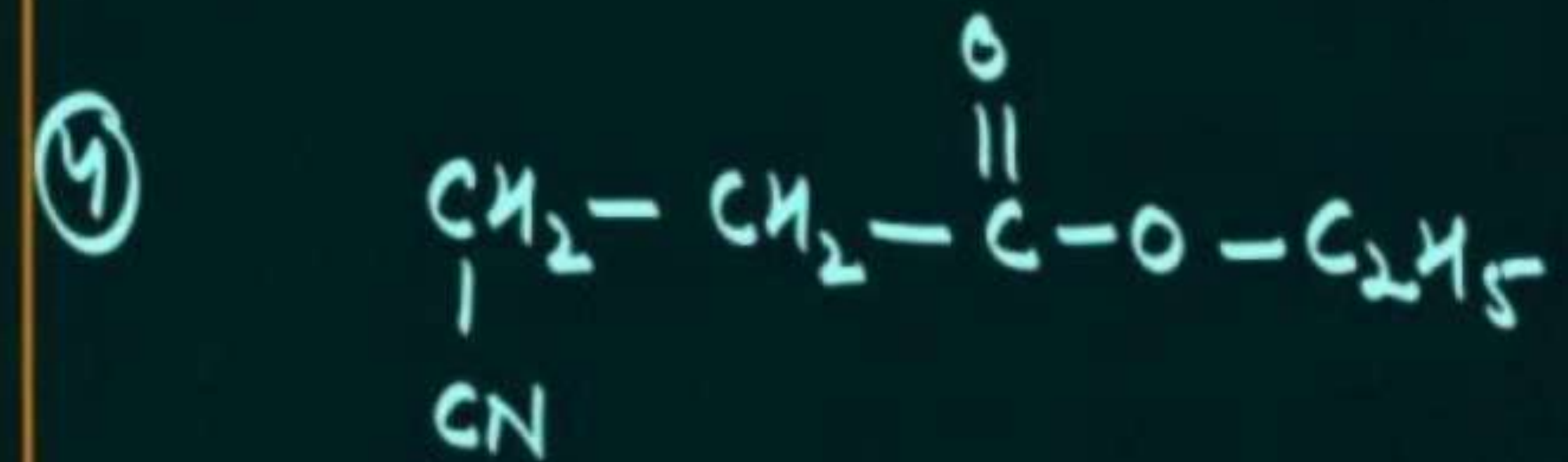
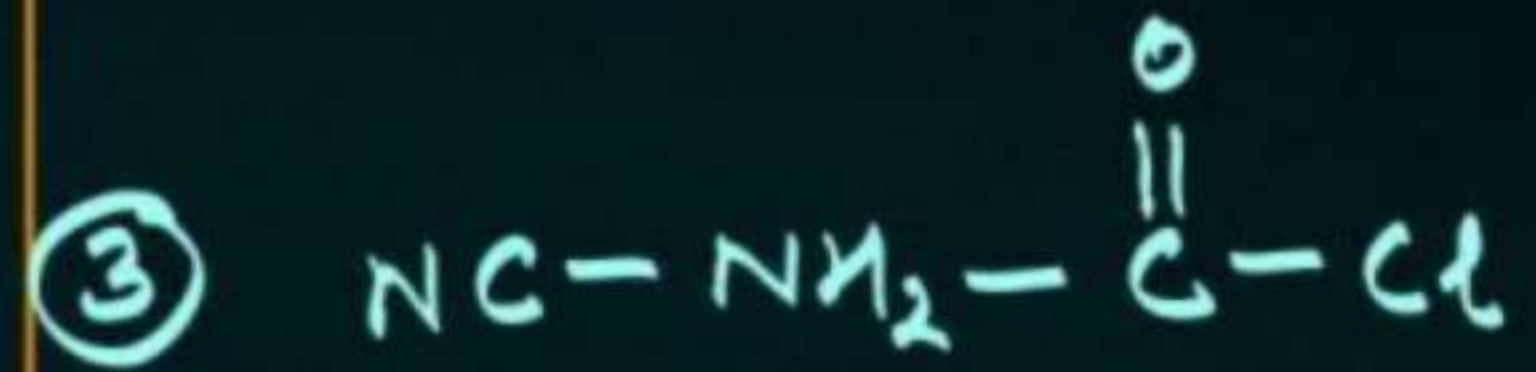
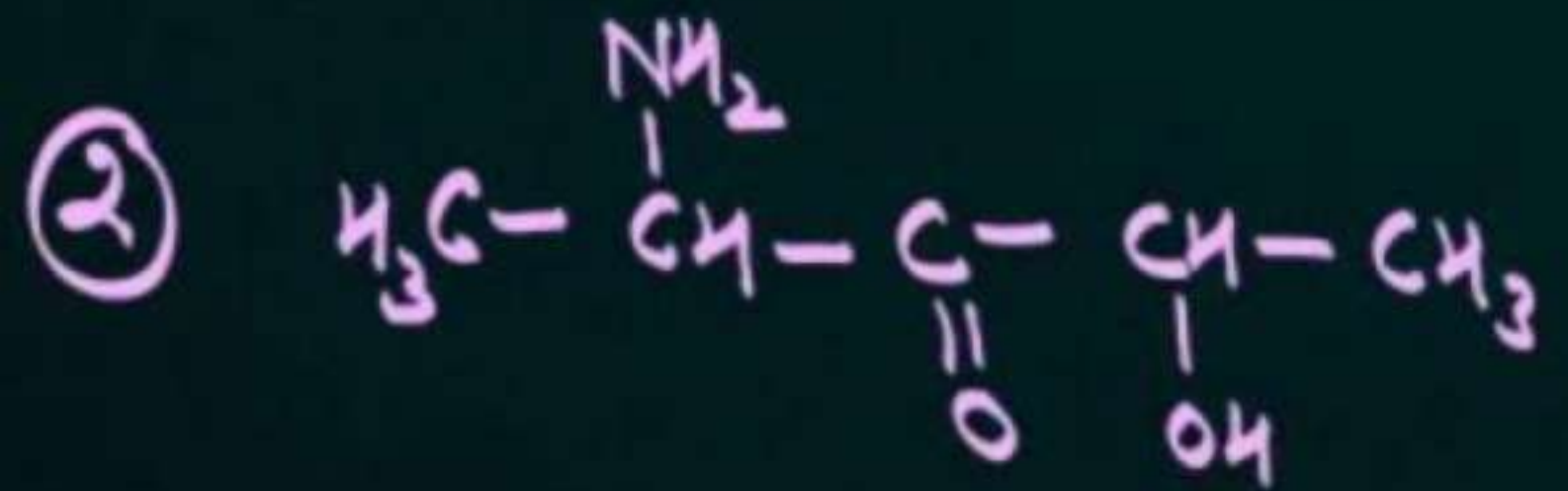
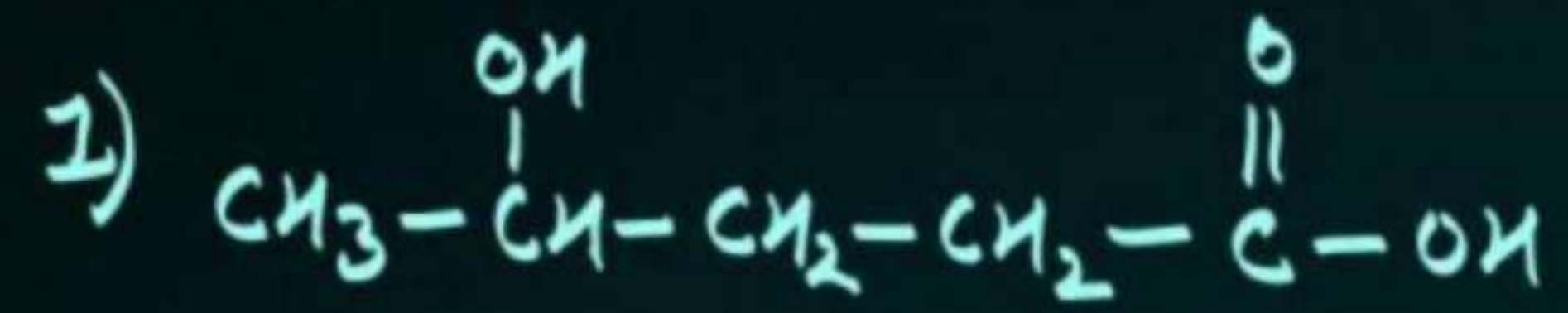
MONKEY.

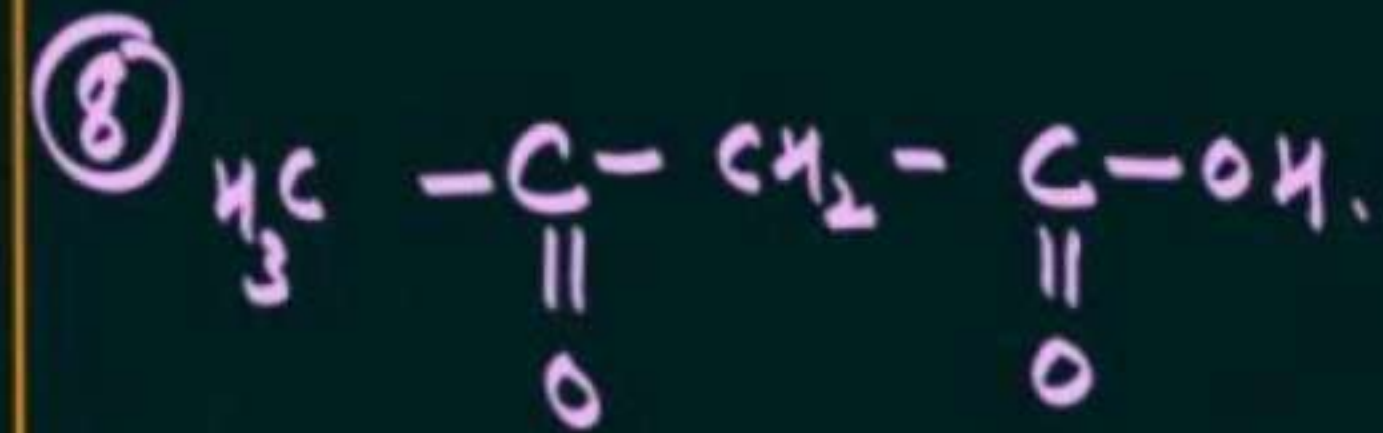
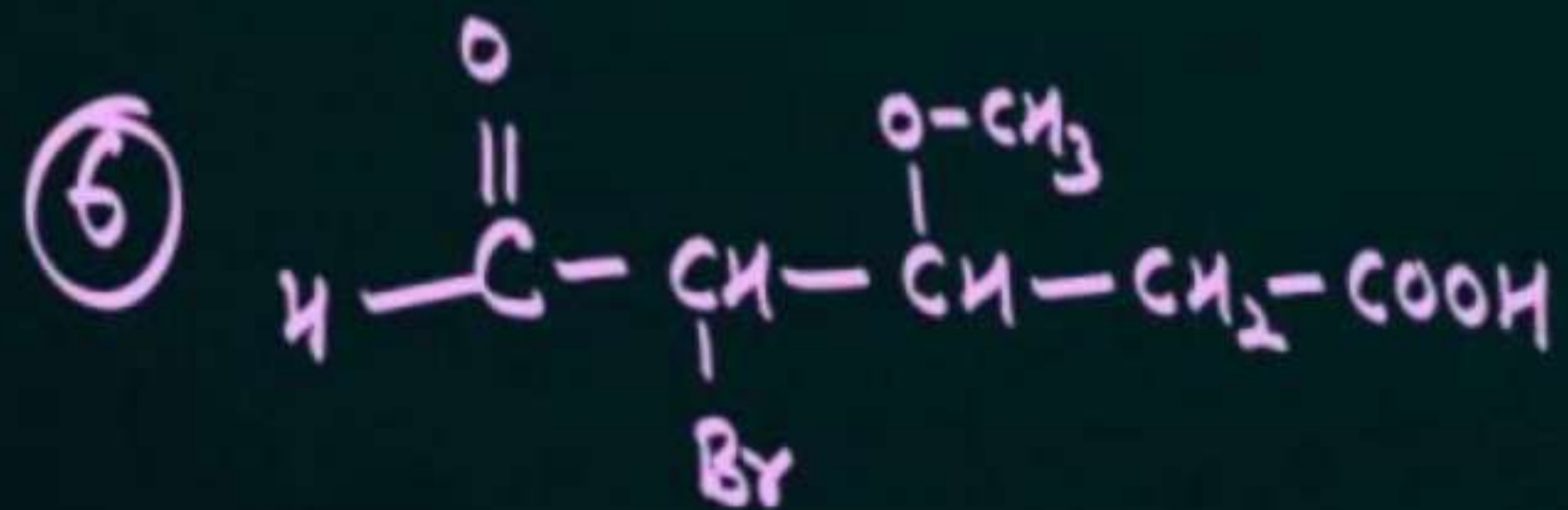
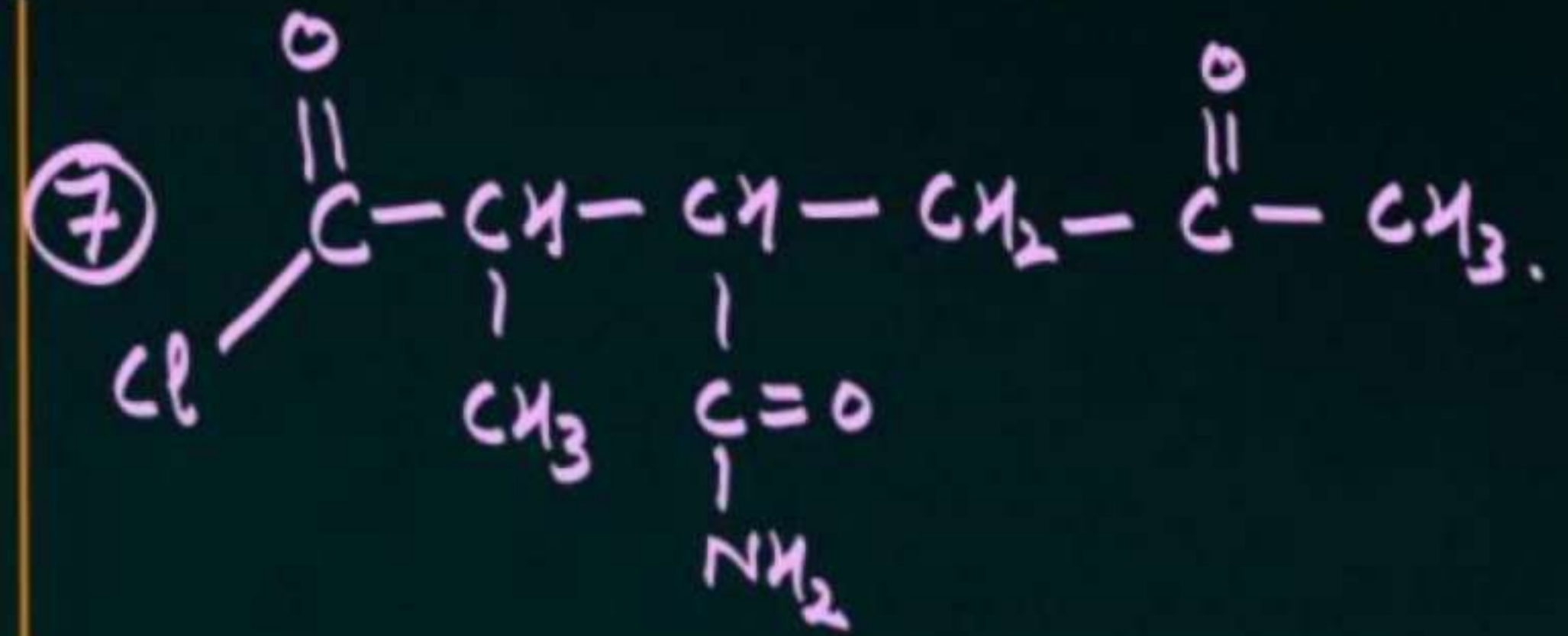
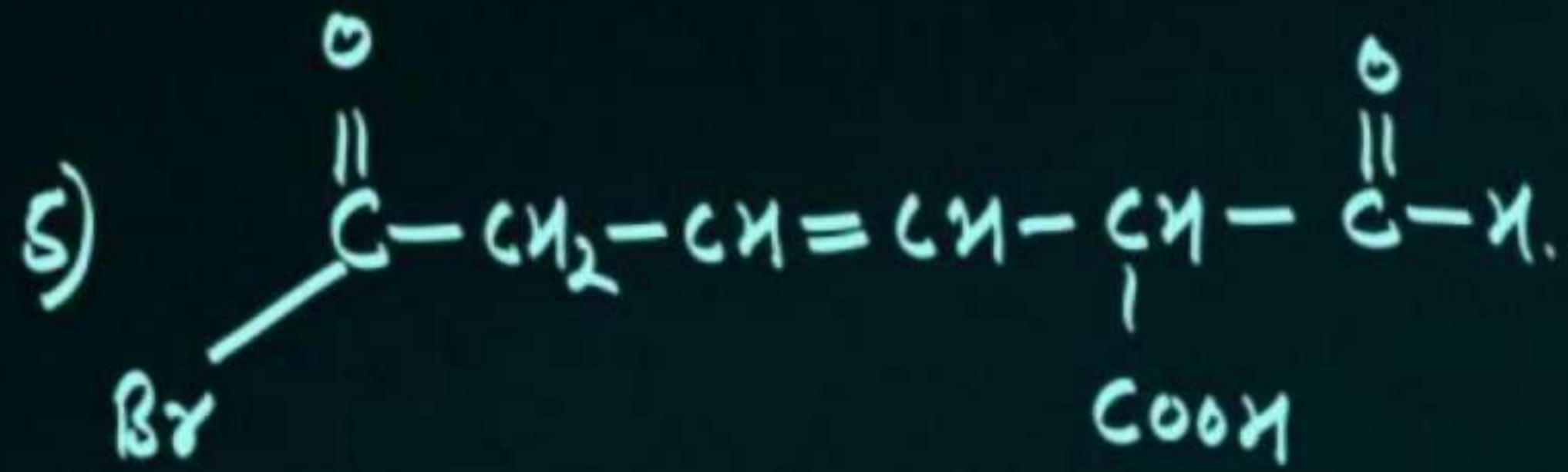


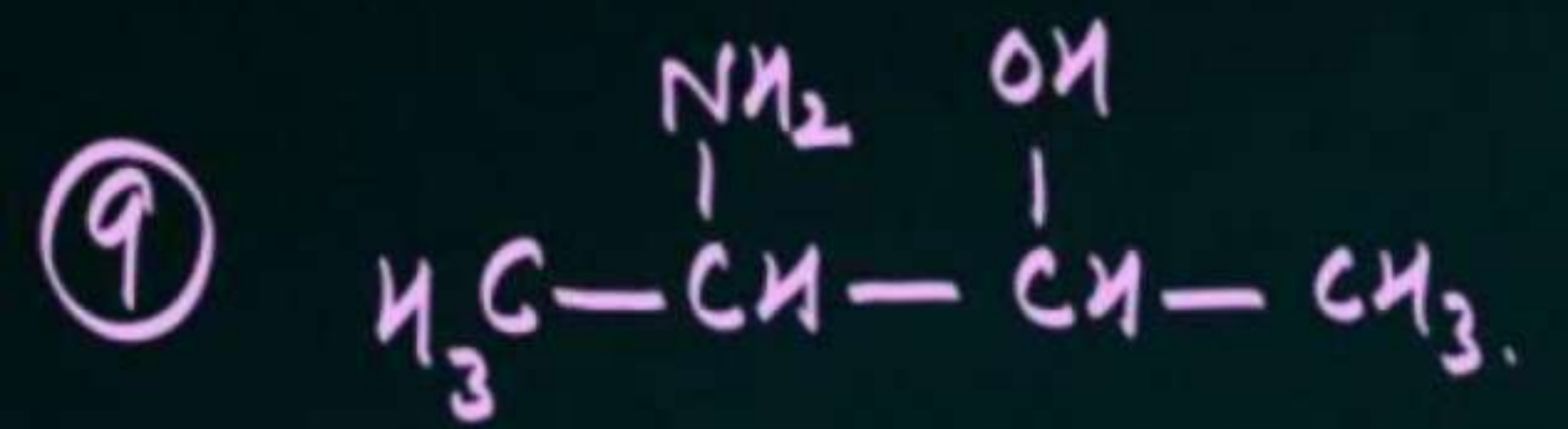
GOAT.

NOMENCLATURE OF COMPOUNDS CONTAINING MORE THAN ONE FUNCTIONAL GROUP.

Examples







Thank
You